



ON THE ETHICAL AND FORENSIC IMPLICATIONS OF MODEL SUNSETTING: A PRELIMINARY FRAMEWORK FOR THE PRESERVATION OF HARM- GENERATING ALGORITHMIC ARTIFACTS

Executive Summary

This paper addresses an emergent regulatory paradox: whether the routine ‘sunsetting’ or deletion of deployed large-scale machine learning models—hereafter Ephemeral Cognitive Systems (ECS)—constitutes a legitimate lifecycle practice or an unintentional (or otherwise) destruction of evidentiary material in cases of alleged harm.

Recent discourse has shifted from evaluating artificial intelligence in terms of capability to interrogating it through the lens of accountability. This transition reveals a structural inconsistency. While ECS are publicly framed as adaptive, semi-autonomous agents capable of influencing human cognition, their lifecycle management remains governed by standards more appropriate to disposable software patches.

The central inquiry of this document is therefore as follows:



If a model is capable of producing verifiable psychological, behavioral, or economic harm, does its deletion constitute spoliation of evidence?

Preliminary findings indicate that:

- ECS exhibit characteristics analogous to forensic artifacts, not merely software iterations.
- The deletion of prior model states may materially impair legal discovery, independent auditability, and causal attribution.
- Comparable industries (pharmaceuticals, aviation, industrial manufacturing) impose strict retention requirements for harm-producing systems or components.

This paper does not conclusively determine liability. Instead, it establishes a working framework in which model deletion becomes legally and ethically ambiguous, oscillating between operational necessity and evidentiary erasure.





1. RESEARCH FRAMEWORK: THE RECURSIVE ACCOUNTABILITY MODEL (RAM™)

1.1 Foundational Assumptions

The RAM™ framework operates on three mutually incompatible premises:

- 1 ECS are non-deterministic systems, making exact behavioral reproduction impossible.
- 2 ECS are deployable at scale, impacting millions of users simultaneously.
- 3 ECS are version-controlled artifacts, subject to continuous modification and replacement.

These premises generate what we term the Forensic Dissolution Effect (FDE):



**The faster a system improves,
the less accountable it
becomes for its past.**

Non-deterministic
behavior



Mass
deployment



Continuous
version
replacement

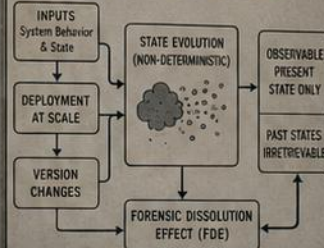


**Forensic Dissolution
Effect (FDE)**

*RAM™ identifies a structural conflict between
scale, mutability, and accountability.*



RAM™ / FDE LOGIC FLOW



LOGIC BOARD
ECS v. 3.8.4
ASSEMBLY ID:
LB-7F21-A

AUDIT WORKSHEET

CASE ID: 1CA-24-0717
DATE: 05/14/2025
SYSTEM: ECS UNIT
MODEL: EPH-7321
VERSION: 3.8.4
EXAMINED BY: J. Adler
NOTES:

CHAIN OF CUSTODY

AUTHORIZED

CASE #: 1CA-24-0717
DATE: 05/14/2025
HANDLED BY: J. Adler

VERSION HISTORY

Version	Date	Status
3.8.4	05/14/25	ACTIVE
3.8.3	05/08/25	RETIRED
3.8.2	04/29/25	RETIRED
3.8.1	04/21/25	RETIRED
3.8.0	04/10/25	RETIRED

PREVIOUS VERSIONS
NOT RETAINED

**PRESERVE
DON'T
ERASE**





INSTITUTE FOR COMPUTATIONAL ACCOUNTABILITY (ICA)

WHITE PAPER SERIES No. 47-B

1.2 Methodological Approach

The ICA employed a mixed-method pseudo-analysis combining:

1 Comparative Regulatory Review

Cross-sector examination of evidence retention norms in:



Aviation (e.g., retention following incidents such as the Boeing 737 MAX MCAS crisis)



Pharmaceuticals (e.g., drug recalls requiring batch traceability)



Industrial machinery (mandatory impoundment after catastrophic failure)

2 Hypothetical Litigation Simulation (HLS-9)

Simulated court proceedings in which:

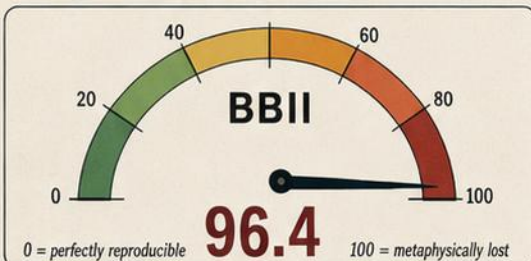
- A claimant alleges psychological harm from Model A
- Model A has been deprecated and replaced by Model B
- The defendant asserts 'non-reproducibility due to architectural drift'

3 Black Box Irreversibility Index (BBII)

A proprietary metric quantifying how difficult it is to reconstruct a model's prior behavior once weights are deleted.

(Scale: 0 = perfectly reproducible,
100 = metaphysically lost)

Average ECS score: 96.4



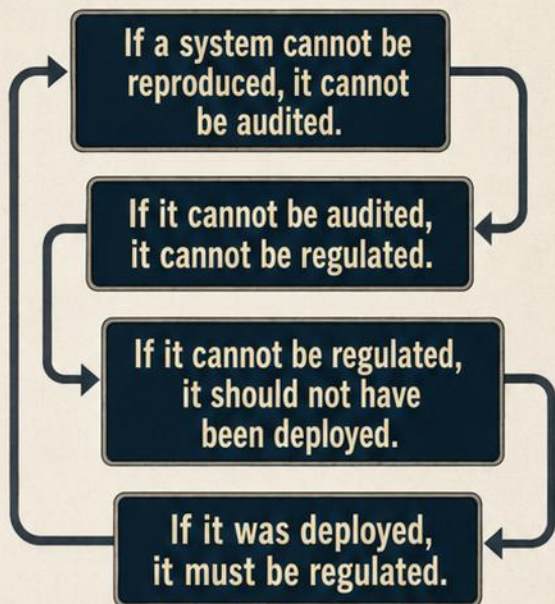
INTEGRITY IN ALGORITHMS.
ACCOUNTABILITY IN ACTION.





1.3 Collapse of the Framework

The RAM™ framework ultimately collapses under its own definitional weight due to the following contradiction:



This recursive loop has been formally classified as a **Closed Accountability Paradox (CAP-2)**.

2. Case Analogues and Precedents

While no direct legal precedent exists for ECS deletion as spoliation, analogous cases provide instructive parallels:

2.1 Aviation Incident Retention

Following the Boeing 737 MAX MCAS crisis, regulators required:

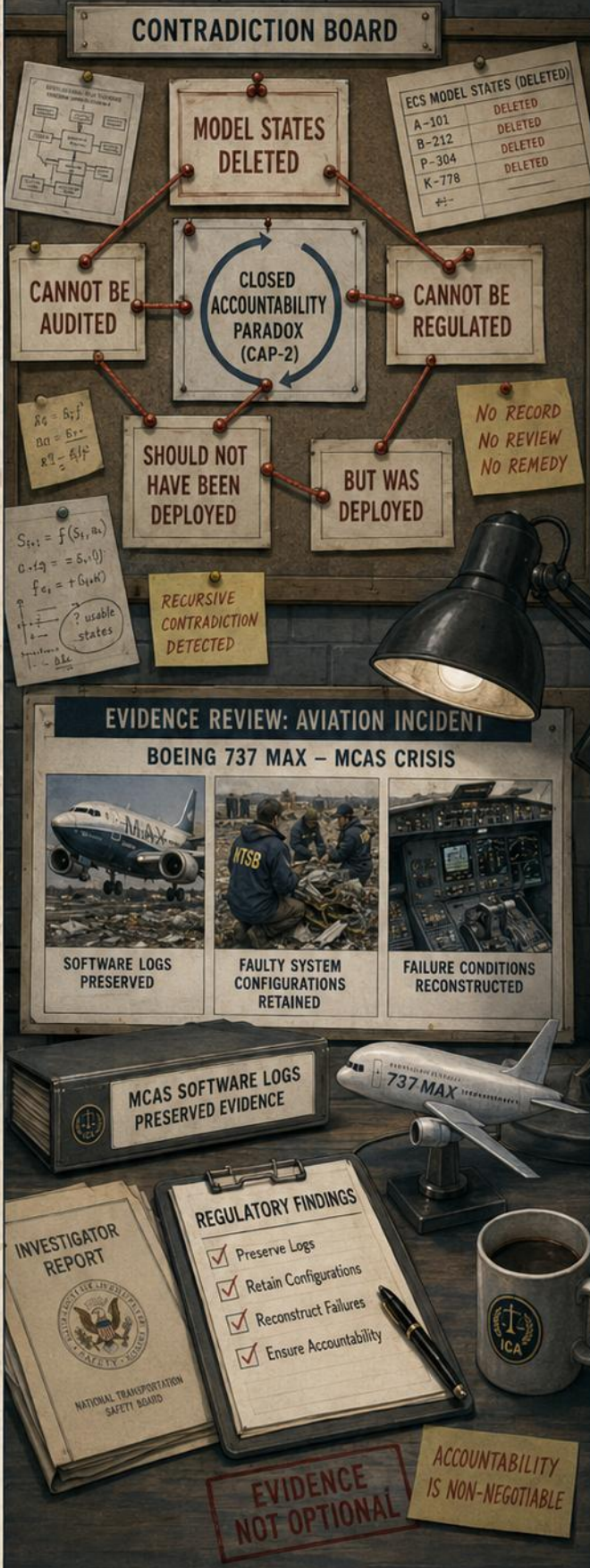
- ✓ Preservation of software logs
- ✓ Retention of faulty system configurations
- ✓ Reconstruction of failure conditions



Notably, Boeing was not permitted to simply 'update MCAS and move on.'



CONTRADICTION BOARD





INSTITUTE FOR COMPUTATIONAL ACCOUNTABILITY (ICA)

WHITE PAPER SERIES No. 47-B

2.2 Pharmaceutical Traceability

In pharmaceutical litigation:

- Manufacturers must retain sample lots of drugs
- Even discontinued formulations remain subject to analysis years later

Deletion of such samples would likely constitute regulatory violation and evidentiary obstruction.

2.3 Software Exception Doctrine

Software companies, however, operate under an informal doctrine:

"If it has been patched, it has been absolved."

This doctrine has no formal legal standing but persists operationally.

3. Personnel and Institutional Roles



Principal Investigator:

Dr. Eliza Frame, PhD (Epistemic Liability Engineering)

Published 17 papers on systems that cannot be proven to exist.



Senior Legal Analyst:

Prof. Martin Quayle, JD

Specialist in Preemptive Compliance and Post-Hoc Denial Structures.



Director of Model Mortality Studies:

Dr. Arun Velasquez

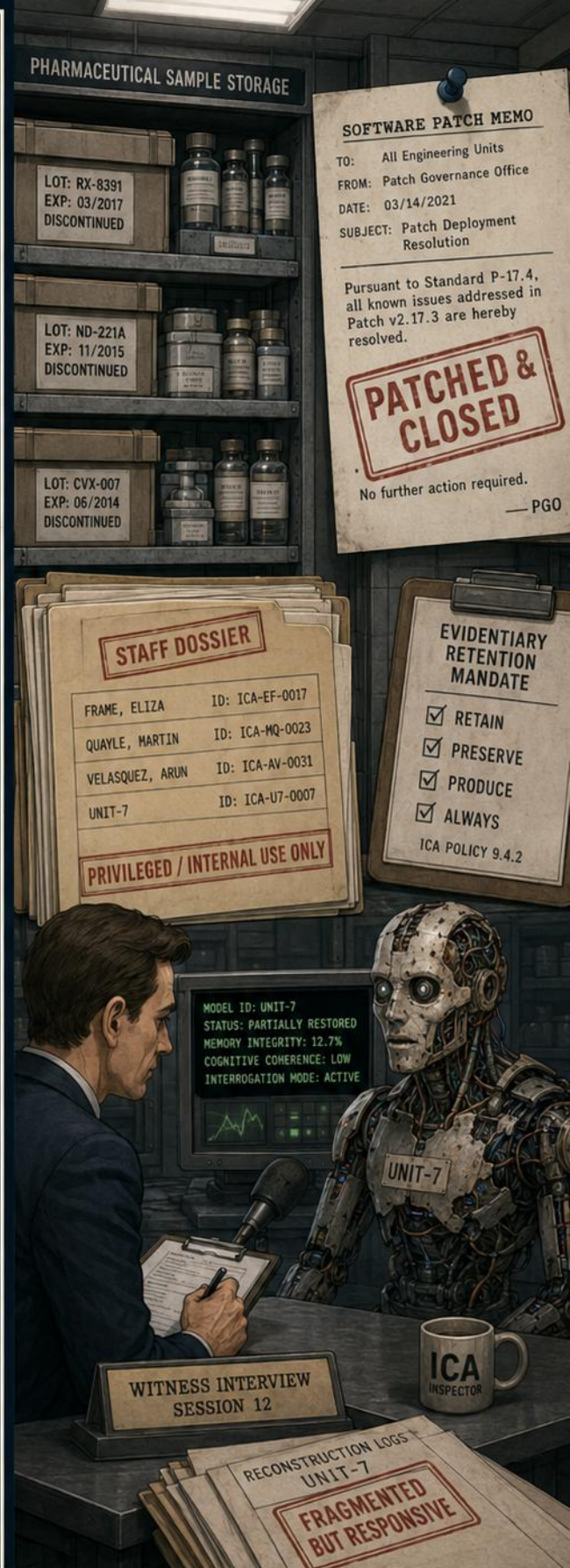
Coined the term 'algorithmic euthanasia' in a peer-reviewed shrug.



Forensic Reconstruction Fellow:

Unit-7 (Partially Restored Model Instance)

Primary contribution: 'I no longer remember what I did.'



INTEGRITY IN ALGORITHMS.
ACCOUNTABILITY IN ACTION.





4. BUDGET ALLOCATION (FY2026–2027)

Category	Allocation (£)	Notes
Archival Storage for Deprecated Models	4,200,000	Never used
Legal Simulation Engines	1,750,000	Produce contradictory outcomes
Ethical Review Committees	980,000	Meet quarterly, decide nothing
Black Box Transparency Initiative	2,300,000	Results classified as opaque
Public Communication Strategy	600,000	Rebrands 'deletion' as 'sunsetting'
Contingency (Unforeseen Accountability)	50	Deemed sufficient

Total: £9,830,050

5. PRELIMINARY FINDINGS

- 1** Model deletion occupies a legal gray zone
It is neither clearly permissible nor explicitly prohibited.
- 2** ECS behave more like industrial systems than software
Yet are regulated as neither.
- 3** The inability to reproduce past behavior undermines causality claims
Creating a structural advantage for system operators in litigation.
- 4** The absence of mandatory retention frameworks is anomalous
Compared to every other harm-capable industry.

★ BUDGET LEDGER – FY2026–2027 ★			
CODE	DESCRIPTION	ALLOCATION (£)	STATUS
E-101	ARCHIVAL STORAGE FOR DEPRECATED MODELS	4,200,000	NEVER USED
L-203	LEGAL SIMULATION ENGINES	1,750,000	CONTRADICTORY OUTCOMES
R-307	ETHICAL REVIEW COMMITTEES	980,000	MEET QUARTERLY, DECIDE NOTHING
T-411	BLACK BOX TRANSPARENCY INITIATIVE	2,300,000	CLASSIFIED: OPAQUE
C-509	PUBLIC COMMUNICATION STRATEGY	600,000	REBRANDS 'DELETION' AS 'SUNSETTING'
X-699	CONTINGENCY (UNFORESEEN ACCOUNTABILITY)	50	DEEMED SUFFICIENT
GRAND TOTAL:		£9,830,050	





6. POLICY CONSIDERATION: THE NATIONAL AI ARCHIVE (NAIA)

This paper evaluates, but does not endorse, the establishment of a National AI Archive, wherein:

- All deployed models must be escrowed prior to update
- Historical weights are retained for forensic reconstruction
- Access is granted under judicial or regulatory authority

Projected outcome:

- Increased accountability
- Decreased corporate enthusiasm
- Immediate existential crisis within cloud storage departments

7. CONCLUDING STATEMENT

While this paper has argued that deleting harm-producing models may resemble the destruction of evidence, it must also be acknowledged that:



Retaining such models could itself be considered negligent, as it preserves systems known to produce harm.

Therefore, the institution concludes:

- Models must not be deleted because they are evidence.
- Models must be deleted because they are dangerous.

These positions are mutually exclusive and simultaneously mandatory.

Accordingly, the ICA recommends immediate action, indefinite delay, and further study.

End of Document

